

BESSEL FUNCTIONS

Bessel's differential equation

Bessel functions, normally labeled J_n , are those obtained by the Frobenius method for solution of the Bessel ODE,

$$x^2 J_n'' + x J_n' + (x^2 - n^2) J_n = 0$$

Generating Function

The functions defined by the generating function are indeed the solutions of the Bessel ODE obtained by the Frobenius method. Generating function formula, a Laurent series, is

$$g(x, t) = e^{(x/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

Rodrigues Formula

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!(n+r)!} \left(\frac{x}{2}\right)^{n+2r}$$

If now we replace n by $-n$, Then

$$J_{-n}(x) = \sum_{r=n}^{\infty} \frac{(-1)^r}{r!(r-n)!} \left(\frac{x}{2}\right)^{-n+2r}$$

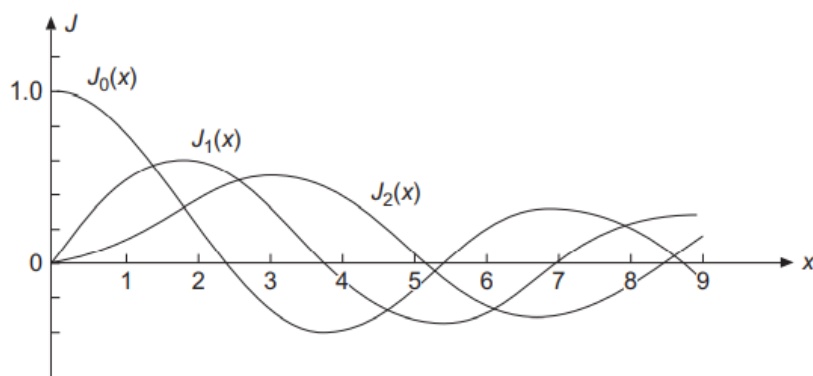
$$J_{-n}(x) = \sum_{r=0}^{\infty} \frac{(-1)^{r+n}}{r!(r+n)!} \left(\frac{x}{2}\right)^{n+2r} = (-1)^n J_n(x) \quad (\text{integral } n)$$

Recurrence Formula

$$\begin{aligned} J_{n-1}(x) + J_{n+1}(x) &= \frac{2n}{x} J_n(x) \\ J_{n-1}(x) - J_{n+1}(x) &= 2J_n'(x) \end{aligned}$$

An important special case is

$$\begin{aligned} J_0'(x) &= -J_1(x) \\ \frac{d}{dx} [x^n J_n(x)] &= x^n J_{n-1}(x) \\ \frac{d}{dx} [x^{-n} J_n(x)] &= -x^{-n} J_{n+1}(x) \\ J_n(x) &= \pm J_{n\pm 1}' + \frac{n \pm 1}{x} J_{n\pm 1}(x) \end{aligned}$$



Bessel – Functions

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

The sinusoidal functions can be written in terms of Bessel's polynomials and they are given by

$$\cos x = J_0(x) + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(x)$$

$$\sin x = 2 \sum_{n=1}^{\infty} (-1)^{n+1} J_{2n+1}(x)$$

Orthogonality

The spherical Bessel functions also possess orthogonality:

$$\int_{-\infty}^{\infty} j_m(x) j_n(x) dx = 0, \quad m \neq n, m, n \geq 0$$

And

$$\int_{-\infty}^{\infty} [j_n(x)]^2 dx = \frac{\pi}{2n+1}$$